

Piecewise Linear Regression

Optimization problem:

$$\min_{\mathbf{x}} f(\mathbf{x}) = \sum_{i=1}^n |a_i \mathbf{x} - b_i|$$

with data pairs $\{(a_i, b_i)\}_{i=1}^n$.

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Equivalent linear programming:

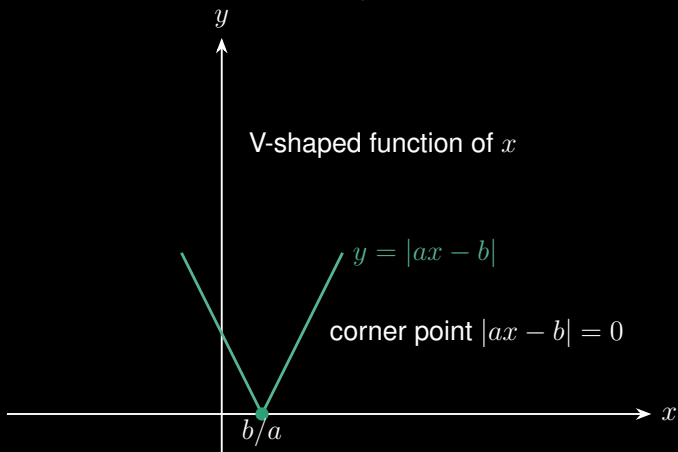
$$\begin{aligned} \min_{\mathbf{x}, t_1, \dots, t_n} \quad & \sum_{i=1}^n t_i \\ \text{s.t.} \quad & -t_i \leq a_i \mathbf{x} - b_i \leq t_i, \quad \forall i \end{aligned}$$

Piecewise Linear Regression

How to minimize $\sum_{i=1}^n |a_i x - b_i|$ without using linear programming?

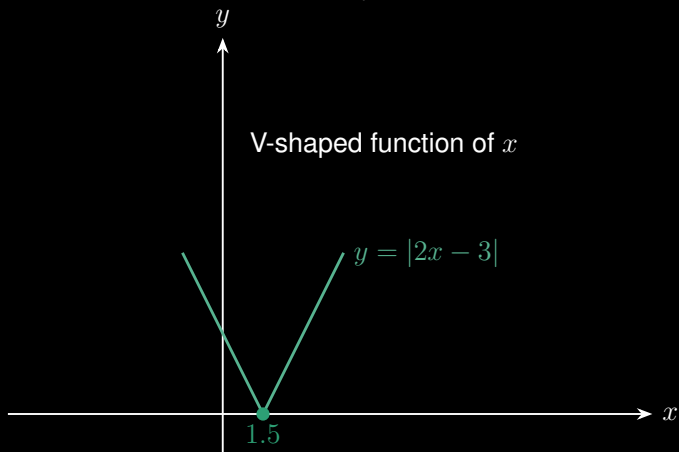
Piecewise Linear Regression

$$\min_x f(x) = \sum_{i=1}^n |a_i x - b_i|$$



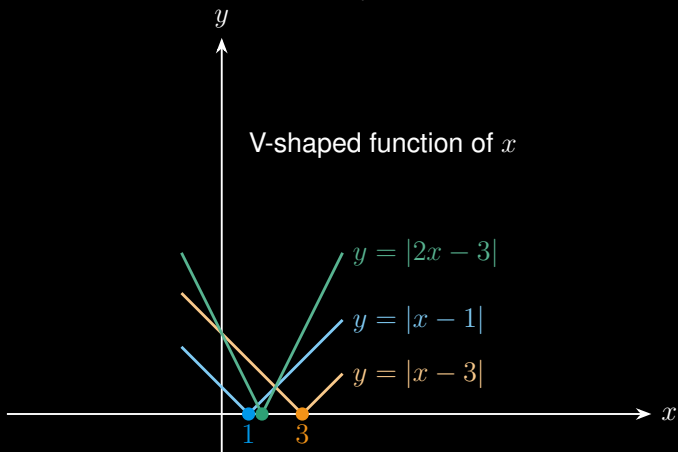
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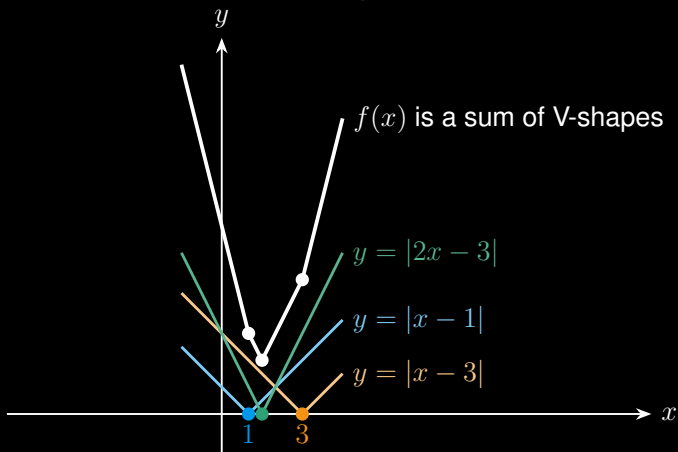
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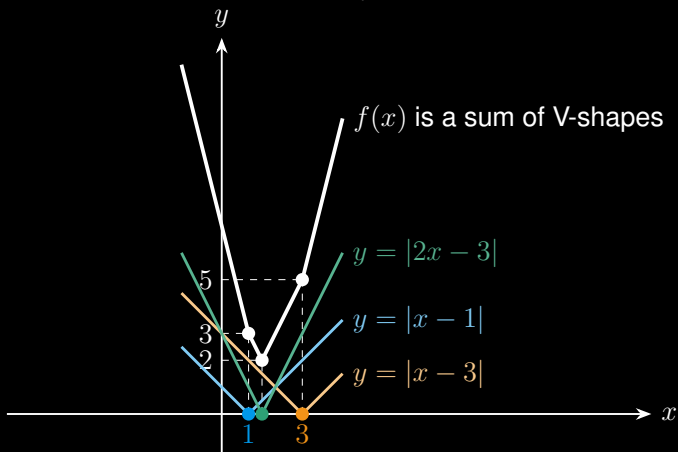
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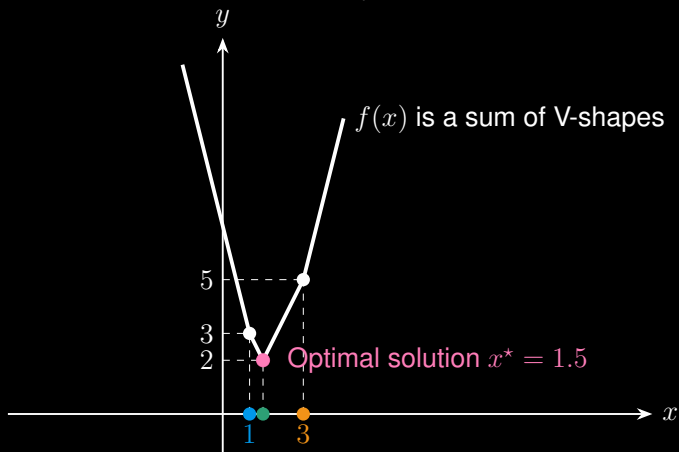
Piecewise Linear Regression

$$\min_x f(x) = \sum_{i=1}^n |a_i x - b_i|$$



Piecewise Linear Regression

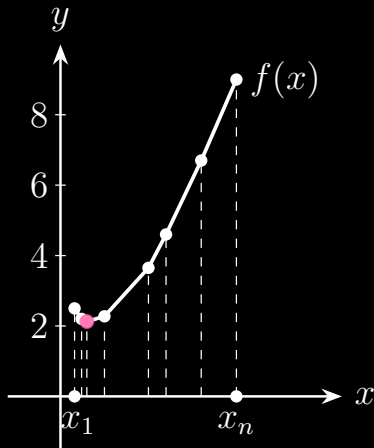
$$\min_x f(x) = \sum_{i=1}^n |a_i x - b_i|$$



Piecewise Linear Regression without Using Linear Programming

$$\min_x f(x) = \sum_{i=1}^n |a_i x - b_i|$$

a_i	b_i	$x_i = b_i/a_i$
0.4	0.5	1.25
0.5	0.2	0.4
0.1	0.3	3
0.2	0.5	2.5
0.4	0.3	0.75
0.1	0.5	5
0.1	0.4	4
0.2	0.5	2.5
0.5	0.3	0.6

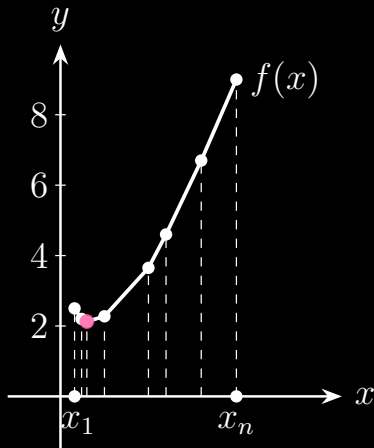


Piecewise Linear Regression without Using Linear Programming

$$\min_x f(x) = \sum_{i=1}^n |a_i x - b_i|$$

Sort values $x_1 \leq x_2 \leq \dots \leq x_n$:

$$x_i \in \{0.4, 0.6, 0.75, 1.25, 2.5, 2.5, 3, 4, 5\}$$



Piecewise Linear Regression without Using Linear Programming

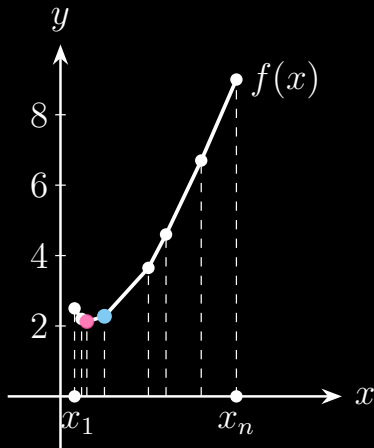
$$\min_x f(x) = \sum_{i=1}^n |a_i x - b_i|$$

Compute $k = \lfloor \frac{n}{2} \rfloor = 4$:

$$x_i \in \{0.4, 0.6, 0.75, \underbrace{1.25}_{x_4}, 2.5, 2.5, 3, 4, 5\}$$

- Positive slope on $[x_4, x_5]$
- Positive slope on $[x_3, x_4]$

$f(x_4)$ is not the minimum value.



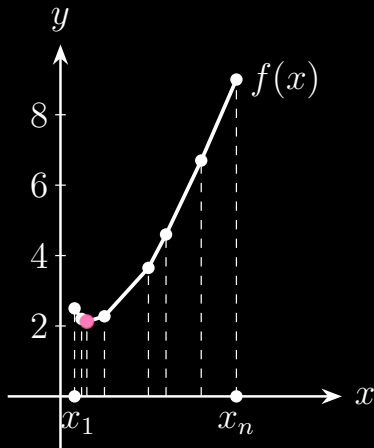
Piecewise Linear Regression without Using Linear Programming

$$\min_x f(x) = \sum_{i=1}^n |a_i x - b_i|$$

$$x_i \in \{0.4, 0.6, \underbrace{0.75}_{x_3}, 1.25, 2.5, 2.5, 3, 4, 5\}$$

- Positive slope on $[x_3, x_4]$
- Negative slope on $[x_2, x_3]$

$f(x_3)$ is the minimum value.



Piecewise Linear Regression without Using Linear Programming

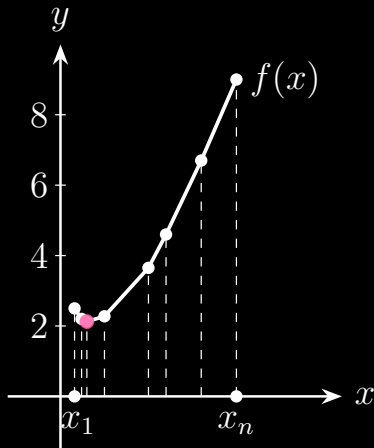
$$\min_x f(x) = \sum_{i=1}^n |a_i x - b_i|$$

Algorithm:

- ① Compute all points $x_i = b_i/a_i$;
- ② Sort values $x_1 \leq x_2 \leq \dots \leq x_n$;
- ③ Let $k = \lfloor \frac{n}{2} \rfloor$ and compute the slope

$$\frac{f(x_{k+1}) - f(x_k)}{x_{k+1} - x_k}$$

- ④ Find some x_ℓ such that the slope is negative on $[x_{\ell-1}, x_\ell]$ but positive on $[x_\ell, x_{\ell+1}]$.



Thanks for your attention!

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